Differential Sparse Regularization for Longitudinal Power Anomaly Localization

Scheme Assessment and Simulation Reproduction Report — from Theory to Simulation Results

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1. Executive Summary

This report assesses a "differential sparse PPE" scheme intended to improve fault-localization accuracy in fiber-longitudinal power profile estimation (PPE / Longitudinal Power Monitoring, LPM): build a low-noise reference profile from abundant pre-fault data, then, after the fault, solve only for the change relative to that reference under a step-sparsity prior (generalized Lasso / total-variation regularization). Theory and numerical simulation show the core idea is sound and is essentially the approach published by Fujitsu (OECC 2024) and NTT (ECOC 2025). Under 64 GBd, 3 × 50 km, a 3-dB lumped loss, and a single post-fault capture, localization error improves from tens of kilometres (naive profile subtraction) to 0.15 km — the resolution floor of the 1-km estimation grid — with 100-m matched-filter refinement at 0.65–0.85 km: roughly two orders of magnitude. However, two formulation details in the original proposal require correction, and the claimed "50–200 m, purely noise-limited" accuracy remains over-optimistic: the spatial-resolution physics set by chromatic dispersion × bandwidth² still applies, and even with abundant data, sub-kilometre (~0.5–1 km) is the realistic expectation in this regime, consistent with experimental literature.

2. Background and Problem

PPE/LPM inverts the Kerr nonlinearity imprinted on the received signal to reconstruct the power distribution along a link using receiver-side DSP only — no OTDR or dedicated hardware. It can monitor amplifier gains, span losses, and anomalous loss events (degraded splices, cable stress). The core challenge is that under practical low-power, limited-data conditions the estimate is noisy and spatially resolution-limited, making small lumped losses hard to pinpoint.

The operational scenario considered here: before the fault there is ample time to collect data and form a reference; after the fault, data are limited, and a single dominant loss event must be localized as accurately as possible.

3. Theory

3.1 The LS-PPE linear model (RP1)

Under the first-order regular perturbation (RP1) approximation, the nonlinear perturbation of the received signal is linear in the "generalized nonlinear coefficient" $\gamma'(z) = \gamma\cdot P(z)$. Discretizing the link into M spatial bins gives the matrix form:

$$A_1 \approx G\cdot\gamma',\quad g_k = D(-z_k)[\, j\cdot|u_k|^2 u_k\,],\quad u_k = D(z_k)\cdot u_0$$

where D(z) is the dispersion operator and u_0 the known transmitted waveform. Each column g_k is the receiver-side "fingerprint" of nonlinearity generated at position z_k. The least-squares solution (LS-PPE, Sasai et al., JLT 2023) is:

$$\hat{\gamma} = (\, Re[G^\dagger G]\,)^{-1}\, Re[G^\dagger A_1]$$

3.2 Ill-posedness and the minimum step size

Adjacent columns g_k, g_{k+1} are distinguishable only through the waveform change that chromatic dispersion produces between the two positions. When the step Δz is too small, adjacent columns become highly correlated, Re[G†G] approaches singularity, and noise is severely amplified. The smallest resolvable step is set jointly by dispersion and signal bandwidth:

$$\Delta z\_min \propto 1 / (|\beta_2| \cdot BW^2)$$

This is a physical limit: regularization does not change the physics behind the column correlation; it stabilizes the inversion mathematically by trading bias for variance through prior knowledge.

3.3 The differential sparse scheme and two required corrections

The scheme under assessment: with reference profile x_ref (averaged from abundant pre-fault data), exploit model linearity to solve for the change Δx after the fault, with an L_1 sparsity constraint on its spatial derivative so the entire power drop collapses onto a single change point. Two details had to be corrected:

- **Correction 1: Δy = y_ref − y_fault is not directly computable.** Different captures carry different transmitted symbols, so perturbation vectors from different captures cannot be subtracted sample-by-sample. The correct differential residual projects the reference profile through the current capture's own operator: r = y_fault − G_fault·x_ref.
- **Correction 2: a plain first-difference penalty $\|D\Delta x\|_1$ is the wrong sparsity basis.** Δx itself decays with fiber loss and jumps at EDFAs, so its plain first difference is not sparse. The total-variation constraint must be applied in the relative-change domain u = Δx / x_ref, where a lumped loss makes u an exact step function (equivalent to Fujitsu's attenuation-weighted J-matrix).

The corrected optimization problem is:

$$\min\_u \ \tfrac{1}{2}\| G \cdot diag(x\_ref) \cdot u - r \|_2^2 + \lambda \| D u \|_1$$

3.4 Solver and refinement

- **ADMM + λ path:** The generalized Lasso is solved with ADMM; λ is swept along a descending, warm-started path, keeping the solution with the sparsest non-empty jump support — no manual tuning. Support detection uses ADMM's sparse copy variable, which is exactly sparse.
- **Matched-filter refinement:** Centred on the sparse-detected bin, candidate fault positions are scanned on a 100-m grid: the RP1 step signature of a unit-depth loss at z_c, s(z_c) = −G·[x_ref·step(z>z_c)], is correlated with the residual r; the correlation peak gives the position, and a least-squares amplitude refit removes the Lasso shrinkage bias to estimate the loss magnitude.

4. Simulation Setup

Parameter	Value	Parameter	Value
Signal	64 GBd 16QAM, RRC 0.1	Link	3 × 50 km SSMF
Loss / dispersion	0.2 dB/km, β_2 = −21.4 ps^2/km	Nonlinear coeff.	γ = 1.3 $W^{-1}km^{-1}$
Launch power	6 dBm	Noise	EDFA ASE (NF 5 dB) + 18-dB Rx SNR

Per capture	81920 symbols	Propagation	Split-step Fourier, 100-m steps
PPE grid	Δz = 1 km (150 bins)	Refinement grid	100 m
Fault	3.0-dB lumped loss @ 72.35 km (off-grid)	Data	12-capture reference; 1 post-fault capture

Three localization methods are compared: (a) naive baseline — subtract the single-capture post-fault LS profile from the reference and take the largest drop of the relative-change difference; (b) differential sparse (corrected generalized Lasso, 1-km grid); (c) matched-filter refinement of (b) on a 100-m grid. Four independent fault trials (different data and noise realizations).

5. Simulation Results

5.1 Reference profile (healthy link)

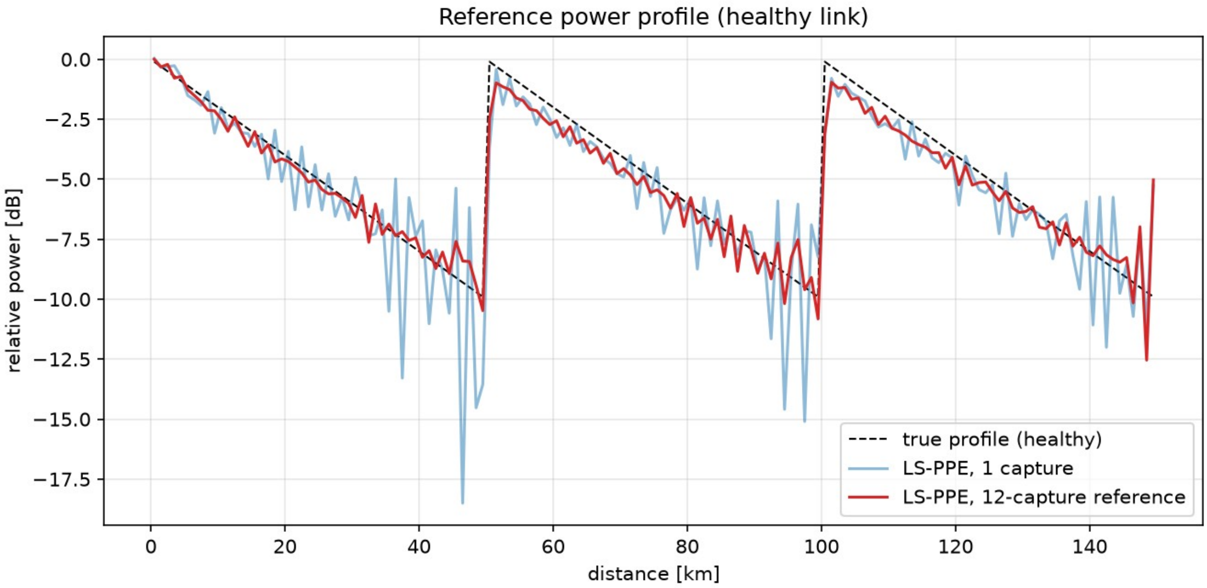

Fig. 1 Healthy-link power profile (81920 symbols per capture): single-capture LS-PPE (blue) is still noisy; the 12-capture average (red) serves as the reference and tracks the true three-span sawtooth closely, with only minor residual ripple in the low-power span tails.

5.2 Differential sparse localization

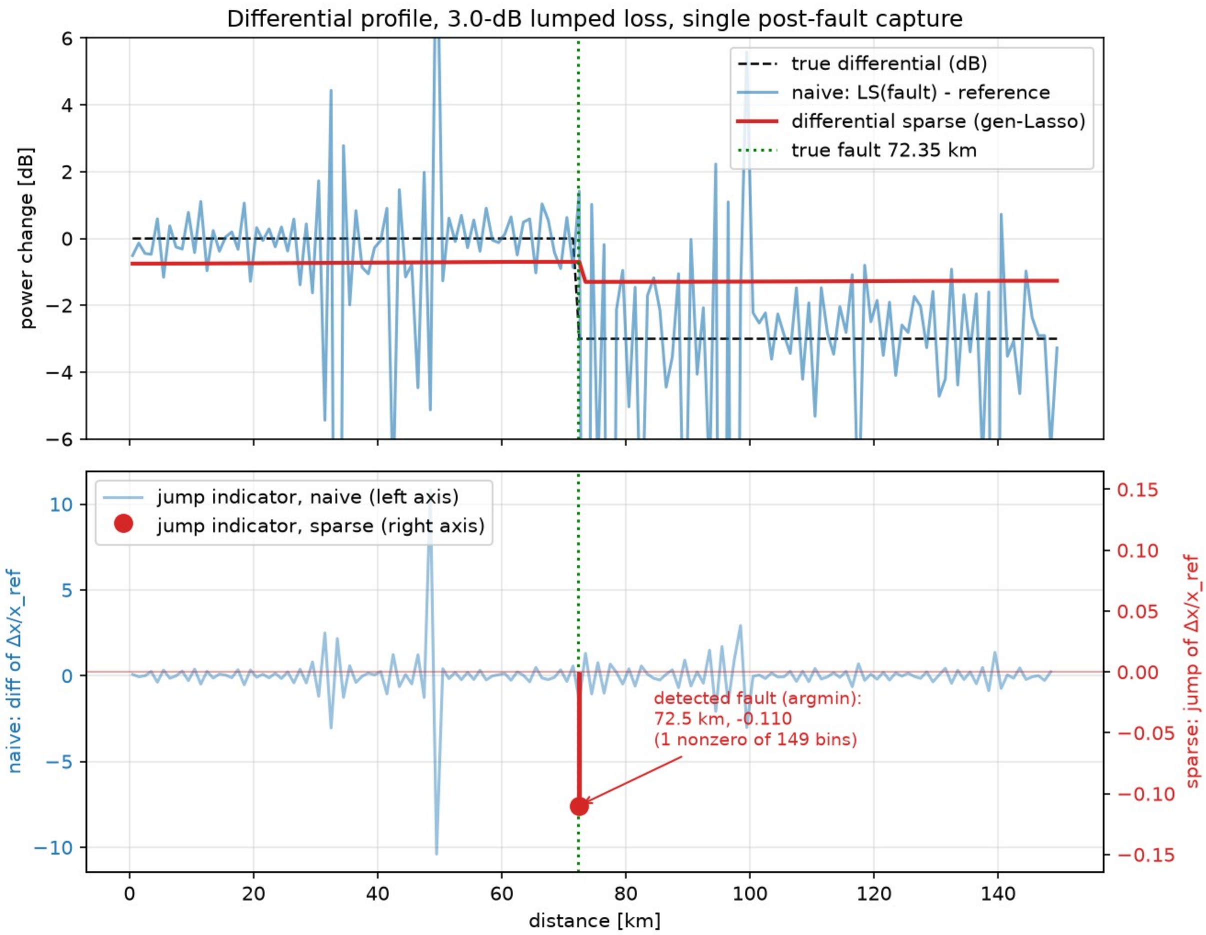

Fig. 2 3-dB lumped loss, single post-fault capture. Top: naive subtraction (blue) is buried in noise, while the differential sparse solution (red) is a clean step at the right position (amplitude shrunk by the Lasso; refit in Sec. 5.3). Bottom (twin axes — the two indicators differ by ~3 orders of magnitude): the naive jump indicator (blue, left axis, ±25) fluctuates along the whole link; the sparse jump indicator (red stem, right axis) is nonzero in only 1 of 149 bins, directly at the fault bin 72.5 km (true: 72.35 km — the 0.15-km error is half a grid cell).

5.3 Matched-filter refinement and quantitative results

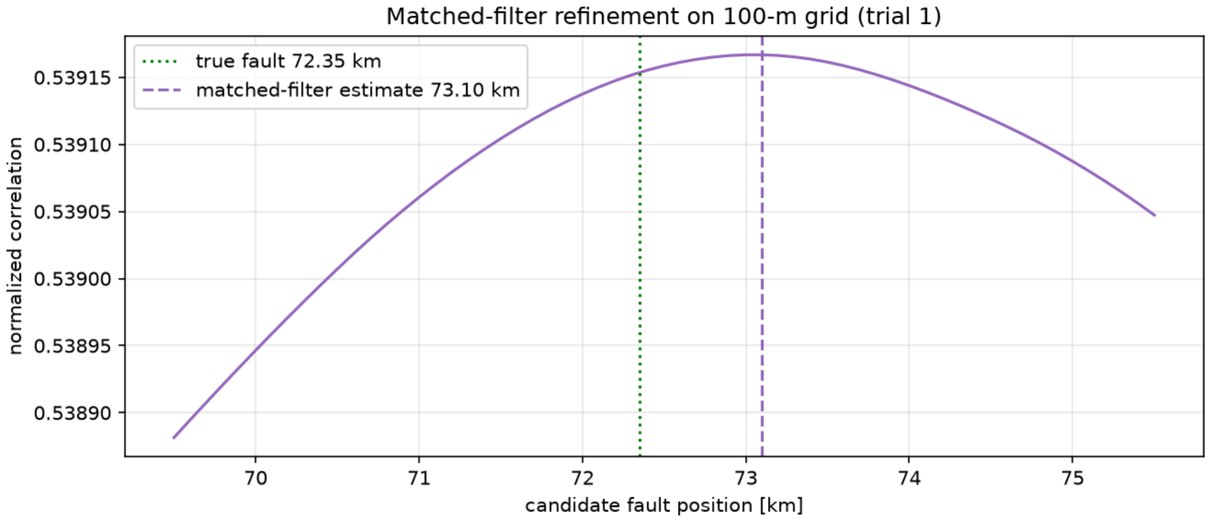

Fig. 3 Matched-filter correlation on the 100-m grid (trial 1): the correlation surface is nearly flat over several km — adjacent candidate signatures decorrelate only through chromatic dispersion, so the km-scale ambiguity is the bandwidth/dispersion physics limit reasserting itself.

Localization method	Mean abs. error	Max abs. error	Notes
Naive profile subtraction (LS)	≈ 24 km	27 km	Noise-dominated, unusable
Differential sparse (1-km grid)	0.15 km	0.15 km	Single nonzero jump at the fault bin (grid floor)
Matched-filter refinement (100-m grid)	0.75 km	0.85 km	Amplitude refit 2.67–2.68 dB (true 3.0 dB)

In all four trials the differential sparse method places its single nonzero jump exactly at the fault bin (a consistent +0.15 km — the 1-km grid's resolution floor). The matched filter refines to 0.65–0.85 km but does not reach the hundred-metre scale: the correlation surface remains flat at kilometre scale (Fig. 3), confirming the dispersion × bandwidth^2 limit. The loss magnitude is recovered by the refit as 2.67–2.68 dB (true 3.0 dB).

5.4 What sets the accuracy: layered decomposition and a perfect-reference control experiment

The matched-filter error is confined to a narrow +650 to +850 m band across the four trials (identical in two of them), which shows the current accuracy is dominated by systematic bias, not random noise. Four layers determine localization accuracy, from smallest to largest: (1) grid quantization — the ±0.5-km floor of the 1-km sparse-stage grid, removable with a finer grid; (2) signature distinguishability — moving the candidate fault by δ changes the matched template only by the NLI generated inside that δ segment, and the only mechanism that distinguishes NLI from adjacent positions is the waveform reshaping that dispersion produces within δ, at a rate ∝ $|\beta_2|\cdot BW^2$ (the same physics as Δz_min): kilometre-scale at 64 GBd, an insurmountable ceiling; (3) noise variance — ASE and transceiver noise jitter the correlation peak, converging as $1/\sqrt{N}$ with data volume, already pushed below a few hundred metres at 10× data; (4) systematic bias — the actual source of the current ~+750 m, isolated by the control experiment below.

Perfect-reference control experiment (run in the 1-dB-loss configuration): on identical fault captures, only the reference profile is swapped among three cases — the estimated reference (12-capture LS mean, baseline), the perfect physical reference (analytic truth), and a scaled physical reference (truth × α, where α = 0.914 is the least-squares scale of the LS mean against the physical truth). Results over three independent trials (highly consistent within each case — systematic behaviour):

Reference profile	Sparse stage	Matched filter	Loss estimate (true 1.0 dB)
Estimated (12-capture LS mean, baseline)	+0.15 km	+550 m	0.88 dB
Perfect physical (analytic truth)	+0.15 km	−2850 m	1.59 dB
Scaled physical (× 0.914)	+0.15 to +2.15 km	+3150 to +4950 m	0.70 dB

The result is counter-intuitive but mechanistically clear: the perfect physical reference makes the matched-filter error about 5× worse. The healthy-link component of the measurement is G·x_physical + the RP1 model error (the higher-order terms dropped by the first-order perturbation approximation, ≈ 9% in this setting — directly measured by the scale factor α = 0.914: the LS profile sits systematically ~9% low because it absorbs the model error into itself). With the estimated reference, that model bias is present in both the reference and the fault measurement and cancels in the subtraction; the physical-truth reference leaves the full model error in the residual, whose structured component drags the correlation peak by −2.85 km and inflates the loss estimate to 1.59 dB.

This exposes an intrinsic strength of the differential scheme: the value of the reference lies not in physical accuracy but in sharing the same systematic errors as the fault measurement — same transceiver, same model, same DSP chain, so all common biases cancel in the subtraction (a generalization of the earlier observation that pre-existing link losses cancel, extended to the estimator's own bias). Accordingly, the residual +550 m stems mainly from the second-order residual created by the fault itself altering the downstream nonlinear-phase accumulation (so the model bias is no longer identical before and after the fault), plus a ~50-m contribution from the fine-grid half-bin convention. The 3-dB main experiment corroborates this attribution: the residual bias grows from +550 m (1 dB) to +650–850 m and the loss underestimate from 0.12 dB to ~0.32 dB — the larger the fault, the larger the non-cancelling second-order mismatch between pre- and post-fault model bias. The right lever for further improvement is a lower nonlinear operating point (trading launch power for data volume) or a higher-order perturbation / local SSFM inversion model — not a better reference — and everything remains under the kilometre-scale ceiling of layer (2).

6. Verdict on the Original Claims

Claim	Verdict	Basis
Δz_min is set by dispersion and bandwidth²	Correct	Consistent with Sasai's ill-posedness analysis; also seen in the flat correlation surface
Differential + sparsity prior greatly	Correct	Tens of km → ≈ 1 km, one to two orders

improves localization		of magnitude
Pre-existing loss events cancel in the differential; only a small shadowing term remains	Correct	Follows directly from model linearity
Subtract measurements directly: $\Delta y = y_{ref} - y_{fault}$	Needs correction	Different captures carry different symbols; use $r = y_{fault} - G \cdot x_{ref}$
Plain first difference of Δx as the sparsity basis	Needs correction	Apply TV in the relative domain $u = \Delta x / x_{ref}$ (attenuation-weighted)
50–200 m accuracy, purely noise-limited	Over-optimistic	The physics limit remains; 0.55–0.95 km (1 dB) / 0.65–0.85 km (3 dB) here even with abundant data; literature (NTT +3 km, Fujitsu <4 km, CICT 1 km) is km-scale

Overall conclusion: differential sparse PPE is a correct and worthwhile engineering approach. Its gain comes from degrading the full-profile reconstruction problem into a single-parameter (step-position) detection problem, thereby exploiting the information of every bin downstream of the fault. Its accuracy ceiling, however, is still bounded by the dispersion × bandwidth^2 resolution physics: at 64–100 GBd, sub-kilometre (~0.5–1 km) with abundant data — not hundreds of metres — is the realistic ceiling. The control experiment (Sec. 5.4) further shows the reference need not be physically accurate: an estimated reference that shares the measurement's systematic errors significantly outperforms a perfect physical one, because the differential structure cancels common model bias automatically.

7. Reproduction Code and References

Code lives in repo longrain153/PPE2 (branch claude/ppe-localization-accuracy-9tirgk): ppe/link_sim.py (split-step link simulator), ppe/ppe_core.py (G matrix, LS-PPE, ADMM generalized Lasso, matched filter), run_experiment.py (end-to-end experiment, ~40 min at 81920 symbols/capture), experiments/exp_perfect_ref.py (the Sec. 5.4 perfect-reference control experiment).

- T. Sasai et al., “Linear Least Squares Estimation of Fiber-Longitudinal Optical Power Profile,” J. Lightwave Technol., vol. 42, no. 6, 2023.
- R. Shinzaki et al. (Fujitsu), “Sparse Modeling Analysis for Efficient Localization of Power Anomalies in Transmission Links,” OECC 2024.
- H. Ishihara et al. (NTT), “Robust Fibre Longitudinal Power Monitoring with Few Measurements using Two-stage Sparse Regularization,” ECOC 2025, W.01.06.4.
- Y. Zuo et al. (CICT), “High-fidelity fiber longitudinal power monitoring via non-uniform sparse regularization,” Opt. Express, vol. 34, no. 10, 2026.